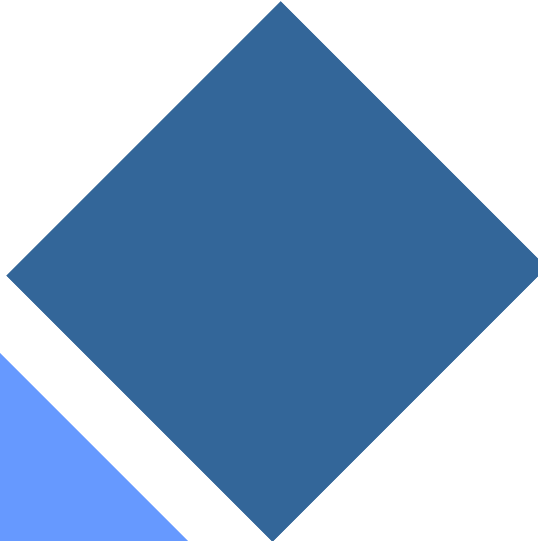


Customer Churn Analysis Project

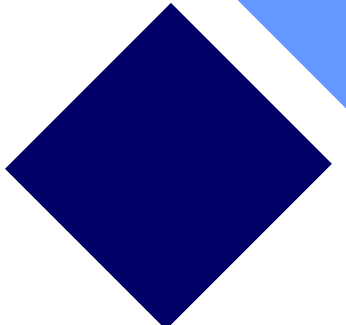
By : Simran Shaw

Github: <https://github.com/simranshaww>





Project Overview



Goal

Analyze customer behavior to identify why customers stop using the company's product/service

Why it matters?

Businesses lose significant revenue when customers churn

Business question

What factors drive churn, and how can it be reduced?

Dataset

- Telco Customer Churn datase
- 7,043 customers, 21 features
- Includes demographics, account details, subscribed services, billing info
- Target Variable: Churn(Yes/No)

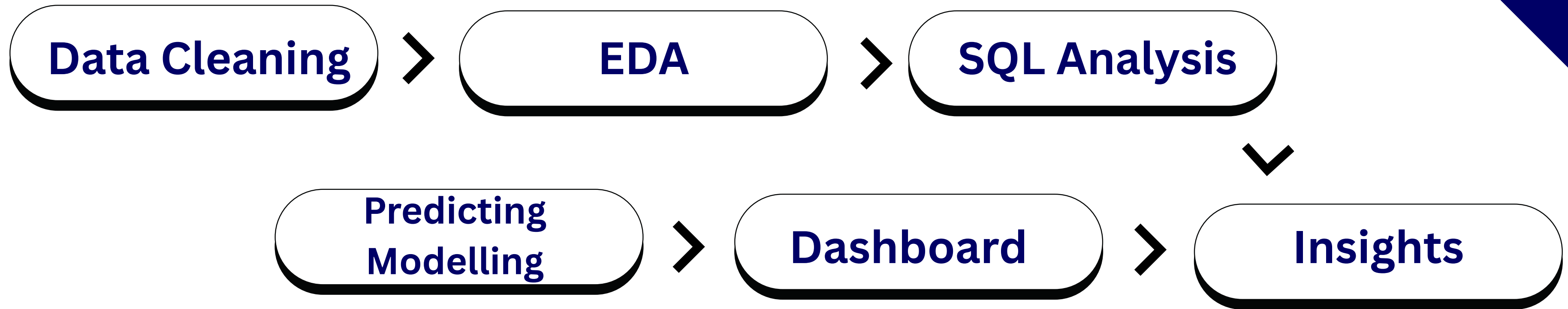


Tools Used:

- SQL (PostgreSQL) – data extraction & analysis
- Excel – data cleaning & preprocessing
- Python (Pandas, NumPy, Matplotlib, Seaborn, scikit-learn) – EDA & modeling
- Power BI – dashboard & visualization



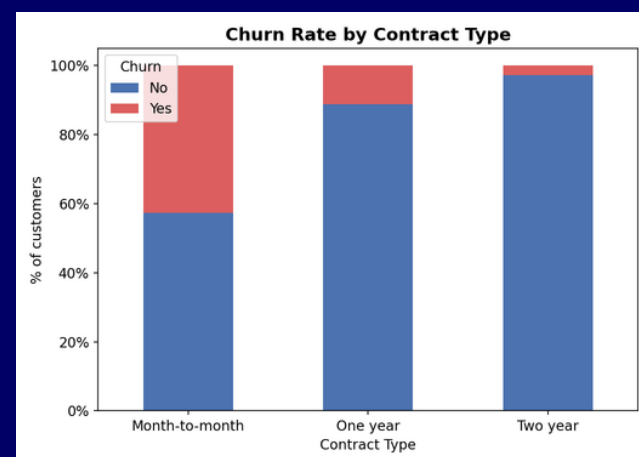
Methodology



Key Insights

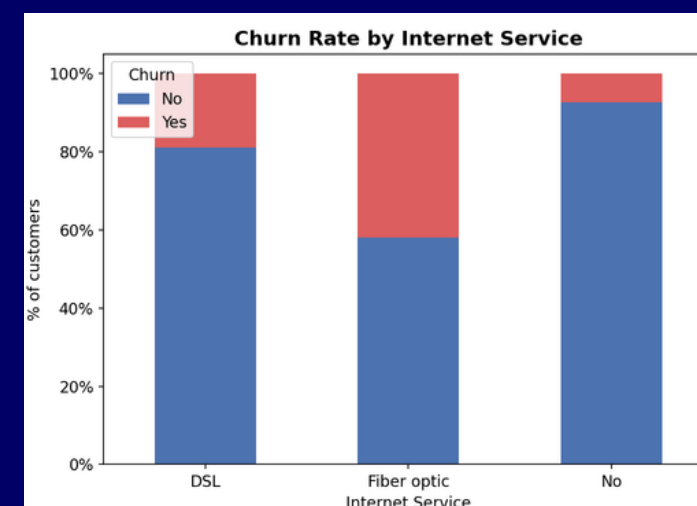
Contract type

- Month-to-month: 42.7% churn
- One-year: 11.3% churn
- Two-year: 2.8% churn



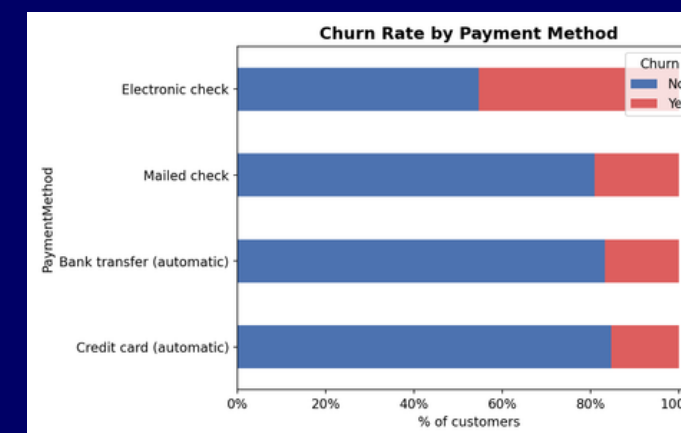
Internet Service

- Fiber optic: 41.9% churn
- DSL: 19% churn
- No internet: 7.4% churn



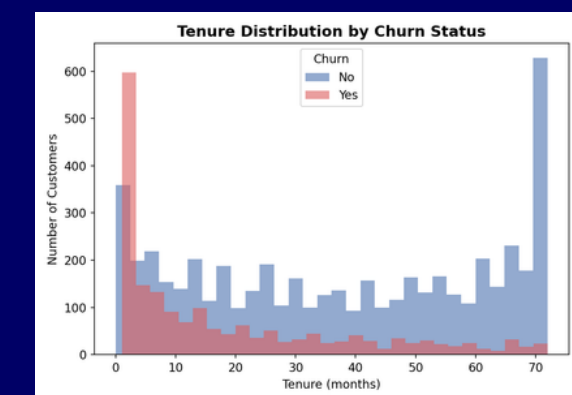
Payment Method

- Electronic check: 45.3% churn (highest)
- Other methods: 15-19% churn



Tenure

- Churned customers: median 10 months tenure
- Retained customers: median 38 months tenure
- Takeaway: churn is heavily front-loaded in the customer lifecycle



Predictive Modeling

~~Logistic regression model built to predict churn

Accuracy : 80%

Top predictors: Contract type, Online Security, Tech Support, Online Backup, Paperless Billing

Note: model catches 56% of actual churners — good for insight, not production-ready



Business Recommendation

- Incentivize longer-term contracts
- Investigate fiber optic price/service quality
- Bundle support services (online security, tech support) into offers
- Focus retention efforts on customer in their first 12 months
- Review electronic check payment method



Interactive Dashboard

CUSTOMER CHURN ANALYSIS DASHBOARD

TOTAL CUSTOMER
7K

CHURN RATE
26.54%

REVENUE LOST
139.13K

Contract

- ☐ Month-to-month
- ☐ One year
- ☐ Two year

Internet S...

- ☐ DSL
- ☐ Fiber optic
- ☐ No

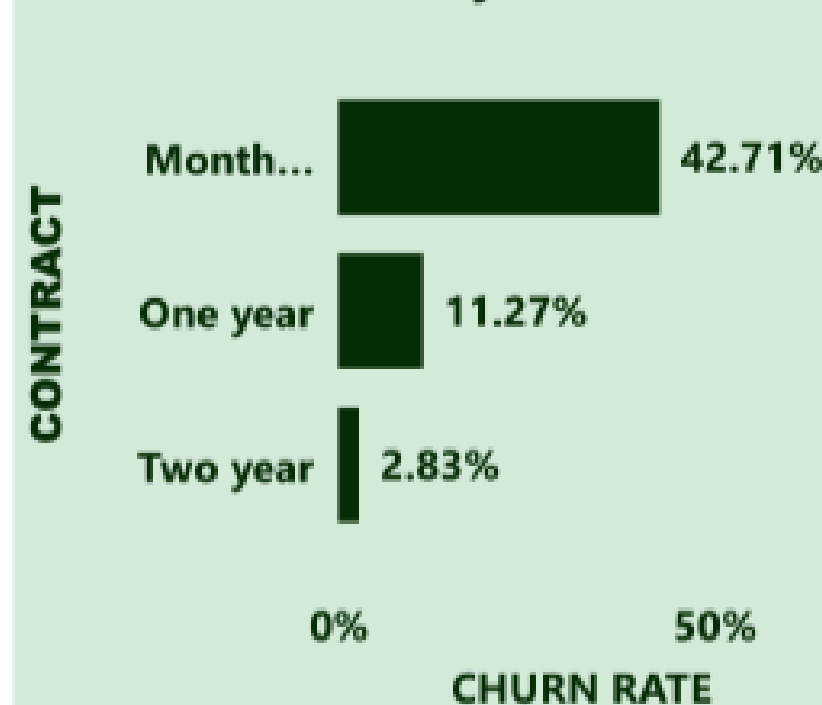
Senior Cit...

- ☐ No
- ☐ Yes

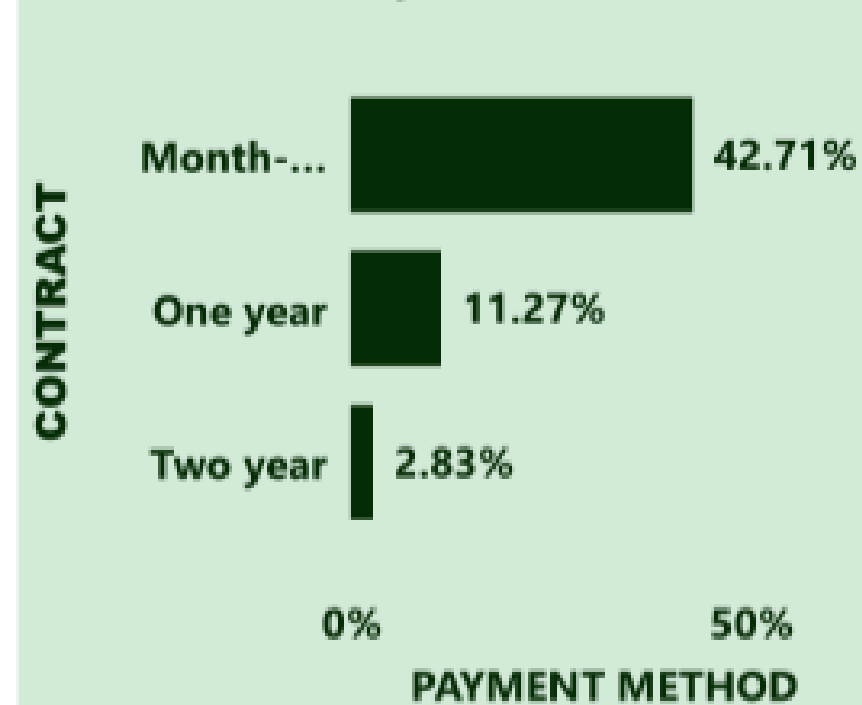
Gender

- ☐ Female
- ☐ Male

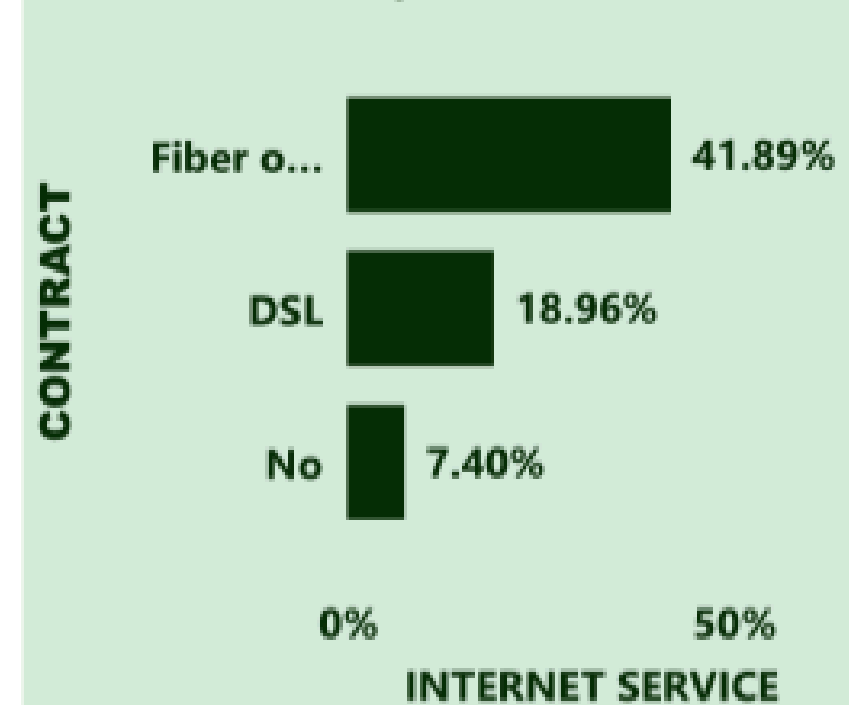
CHURN RATE by CONTRACT



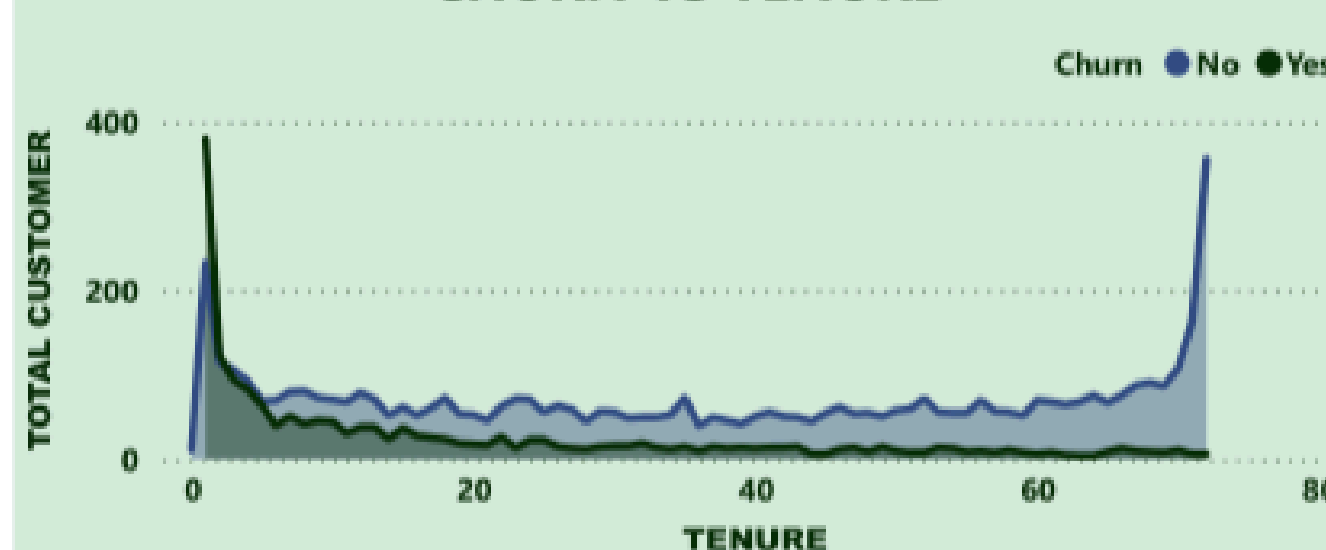
CHURN RATE by PAYMENT METHOD



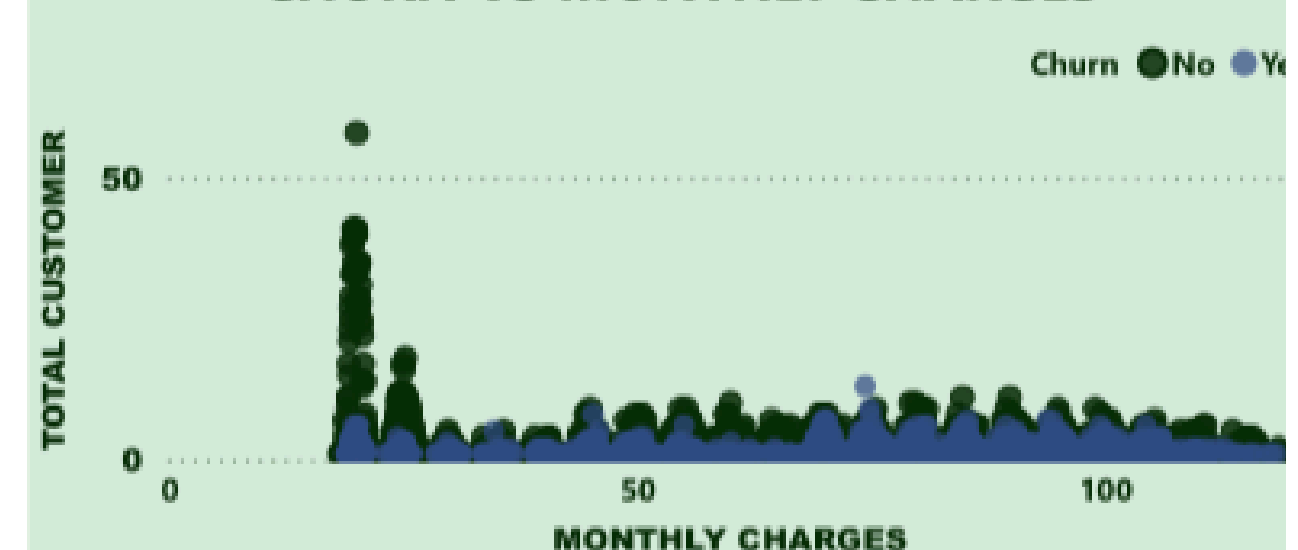
CHURN RATE by INTERNET SERVICE



CHURN VS TENURE



CHURN VS MONTHLY CHARGES



Conclusion

The overall analysis concludes - Contract types, tenure and service add ons to be the strongest churn predictors in Telco Customer data. These insights can directly inform retention campaigns and reduce revenue loss.

Thank
you

